HateReasoner: Detecting Hateful Memes via Presupposed Context Modeling and LLM-Guided Graph Reasoning

Q.Q. Chen (2330034001), S.P. Chen (2330026016), G.C. Jiang (2330034021)
M.H. Li (2330034027), Y.R. Liang (2330034030), Y.X. Tu (2330034048)
Beijing Normal-Hong Kong Baptist University

***Abstract*—Automated hateful meme detection is a crucial yet highly challenging task. Unlike traditional text-based hate speech, the offensiveness of a meme often does not reside solely in the image or the text, but emerges from their combination, generating implicit meanings such as sarcasm, factual distortion, or reinforcement of group stereotypes. Existing multimodal methods often struggle to effectively capture these complex signals, which rely on social common sense and contextual reasoning. To address this, this paper proposes a novel framework named HateReasoner, which performs joint reasoning through three complementary semantic pathways: a Presupposed Context Module (PCM) that explicitly models strong interactions between the image and text, a Semantic-Context Fusion Module (SCFM) integrates a Semantic Guidance Module (SGM) and a Graph-based Multimodal Module (GMM). The former leverages large language models to provide high-level semantic guidance for recognizing sarcasm, misinformation, and implicit accusations, while the latter models complex relationships between local image and text elements to achieve deeper semantic fusion.. Experimental results demonstrate that the proposed method significantly outperforms existing state-of-the-art models.**

***Index Terms*—Hateful Meme Detection, Multimodal Detection, Semantic Reasoning, Graph Neural Network**

I. INTRODUCTION

With the rapid advancements in internet and photo-editing technologies, memes have emerged as a prevalent form of communication on social media. While the majority of memes are innocuous, some may include harmful or hateful imagery or text. On an individual level, studies have shown that exposure to negative memes can have detrimental effects on mental health (1). On a broader societal scale, memes that promote racism or secessionism can exacerbate social divisions. Furthermore, research indicates a potential link between such memes and real-world criminal activities (2).

Unlike traditional text-based hate speech, hateful memes are inherently multimodal. Their offensiveness often only becomes clear when the image and the text are interpreted together. Text or images that appear harmless on their own can convey strong derogatory or attacking messages when combined. These meanings often rely on sarcasm, implicit stereotypes, or distorted facts.

In this sense, a hateful meme can be understood as a combination of image, text, implied meaning, and rhetorical devices, such as irony, insinuation, or misinformation. Because hateful intent emerges across modalities and depends heavily on context, detecting hateful memes is far more complex than traditional single-modality hate speech classification.

Most existing approaches focus on either textual natural language processing or visual recognition models. In recent years, CLIP-based multimodal embedding methods have been proposed to project images and text into a shared semantic space. However, these methods mainly capture surface-level alignments and often fail to understand the implicit social, cultural, and factual knowledge embedded in hateful memes. For example, a neutral sentence may reinforce group stereotypes when paired with a specific image. Similarly, an image that seems harmless may convey stigmatizing or misleading messages when combined with suggestive text. In addition, some memes deliberately distort factual information. Therefore, an effective hateful meme detection system must go beyond simple image–text alignment. It needs to reason about latent semantic relations, social commonsense, and potential misinformation, which can lead to malicious or ideological hate.

To address these challenges, we propose a CLIP-based multimodal hateful meme detection framework that integrates three complementary semantic pathways. First, the Presupposed Context Module (PCM) models image–text interactions under the assumption of strong contextual dependency. It highlights semantic features that only emerge when both modalities are considered together. Second, the Semantic Guidance Module (SGM) leverages large language models (LLMs) to provide high-level semantic guidance. This module helps identify misleading claims, implicit accusations, and sarcastic expressions that may not be explicit in the original image or text. Finally, the Graph-based Multimodal Module (GMM) constructs a graph over image patches and text tokens. A graph neural network is then used to model both intra-modal and cross-modal relations, enabling deeper contextual reasoning. Through joint learning across these three pathways, the proposed framework captures not only explicit image–text correspondences but also hidden semantic signals. As a result, it significantly improves the accuracy of hateful meme detection.

II. LITERATURE REVIEW

A. Image inpainting

Most existing multimodal hateful meme detection approaches rely on masked images combined with textual input for joint understanding. A representative example is HateCLIPper (3), which directly feeds masked visual content and associated text into a CLIP-based framework for classification. While such designs aim to reduce noise or emphasize salient regions, they introduce a fundamental limitation: the model is forced to reason over visually incomplete scenes, which biases learning toward local and fragmented correspondences rather than a holistic understanding of global image semantics and contextual coherence.

When visual understanding is restricted to masked regions, models are likely to miss hateful logic that only emerges through the integration of multiple visual cues, including human posture, environmental context, and socially meaningful symbols (e.g., uniforms, skin color), as well as their interaction with the accompanying text under a presupposed contextual frame. Consequently, hate expressions that depend on global scene composition and socio-cultural interpretation may remain undetected.

By contrast, restoring masked regions through *image inpainting* enables the model to perceive a more complete and semantically coherent visual scene, which is essential for accurately understanding complex hate expressions that rely on contextual and relational cues.

In this line of work, the Big LaMa framework (4) applies large-mask inpainting using Fourier Convolutions (FFC) and global receptive fields, allowing missing regions to be reconstructed with coherent structure, semantic consistency, and smooth texture transitions even under heavy occlusion.

By preserving global context while recovering fine-grained details, such inpainting methods are particularly well suited for meme preprocessing, where overlaid text regions are often large, irregular, and semantically entangled with the visual scene.

B. Cross-modal understanding of images and text

Early approaches primarily relied on joint feature learning or attention-based mechanisms to align visual and textual representations, enabling tasks such as image–text retrieval, visual question answering, and content generation. A representative example is the ALFRED system proposed by Zhu (5), which enhances multimodal transformers through external labels and in-domain pretraining for hateful meme classification.



Fig. 1: Examples of images in Hateful Memes Challenge

However, such methods are often constrained by the representational capacity of single-modality features, making it difficult to capture high-level semantic relationships between images and text.

With the advent of contrastive learning, Contrastive Language–Image Pre-training (CLIP) leverages large-scale image–text pairs to map visual and textual data into a shared semantic space, enabling more effective cross-modal alignment and semantic retrieval. Nevertheless, relying solely on CLIP-based alignment remains insufficient for capturing implicit semantics and social commonsense knowledge, which are critical for understanding hateful memes. As a result, it is necessary to further incorporate semantic priors and relational reasoning modules to achieve a comprehensive understanding of complex hateful logic.

C. Graph-based Reasoning in Multimodal Understanding

Beyond feature alignment and attention-based fusion, we further reviewed graph-based formulations for multimodal reasoning. In particular, Li et al. (6) propose a visual semantic reasoning framework that represents image regions and textual elements within a unified graph structure, enabling relational reasoning for image–text matching. Their work demonstrates that explicitly modeling interactions between visual and linguistic components allows the model to capture higher-level semantic dependencies that are difficult to obtain through feature alignment alone.

This line of research suggests that multimodal understanding can benefit from treating visual and textual units as interdependent entities connected through structured relations, rather than as independently encoded features.

Inspired by this insight, we observe that hateful meme detection similarly requires reasoning over the relationships between image regions and textual tokens, especially when hateful meaning arises from their joint interpretation. This motivates the incorporation of graph-based reasoning into our processing pipeline, where image patches and text tokens are jointly modeled as a multimodal graph to support structured inference prior to final classification.

III. METHODOLOGY & IMPLEMENTATION

A. Data Preprocessing

In order to make sure that the model works well for recognizing memes, we did some organized work on the Hateful Meme Challenge dataset before using it. The dataset consists of around 10,000 meme samples, with each sample containing an image and corresponding text. Textual information might be included in the pictures, so we have to process all kinds of data together.

1) Text Extraction

To extract embedded text from meme images, we employ Optical Character Recognition (OCR) technology. Given an input image $I \in \mathbb{R}^{3\times H\times W}$, the OCR model $\mathcal{O}$ maps it to a recognized text sequence:

$$T = \mathcal{O}(I) = \{t_1, t_2, \ldots, t_L\}, \tag{1}$$

where L denotes the number of recognized text tokens, and t_i represents the i-th token. This step retrieves visible textual information from the image, providing essential input for subsequent semantic analysis.

During text extraction, the OCR system also supplies location coordinates (x_i, y_i, w_i, h_i) for each token t_i. This spatial information of text regions ensures that subsequent processing stages can accurately localize text within the image.

2) Mask Generation

Extracted texts need to be masked after being extracted so that it will not interfere with the subsequent operation of the model to extract the texts of the image. In this research, we use a mask generation method based on Efficient and Accurate Scene Text Detector (EAST) for the text areas on the meme images (7). It removes obvious text interference from the picture restoration and meaning analysis process, but still keeps most of the hidden picture information.

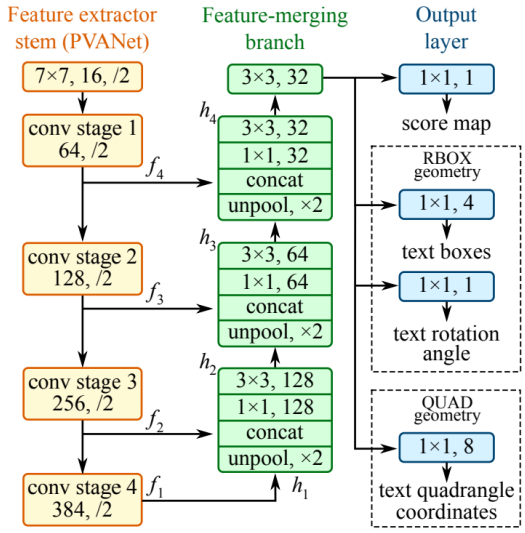


Fig. 2: The Structure of EAST model (7)

Firstly, the input images are all uniformly resized to a certain size so that the detection network can have a unified input scale. Image scaling ratio is calculated for later use to map the predicted text boxes back to the original image coordinates.

Next, forward inference with the EAST model gives us text confidence scores and related geometric info for every pixel, creating candidate text bounding boxes. For line level alignment, group all the textboxes on one line by their center y coordinate and their average height to get a better line level alignment. Then we combine the min and max coordinate of each line's text box to form the final mask boundary.

During mask generation, gray rectangles are overlaid on the corresponding positions of each text region in the original image to create the masked image.

This mask can remove most of the text information on memes, so that the model will not be disturbed by text when extracting visual features later. It gives dependable input for later inpainting or multi-mode semantic alignment work.

3) Image restoration with Big-LaMa

After extracting texts and generating masks, we use Big-LaMa model to do image inpainting on meme images. The purpose is to recover the original image before the text is overlaid so as to extract visual features better. Therefore, it can combine local features with surrounding contexts for a comprehensive judgment.

To get all the spatial and frequency domain info out of the picture, we first split up the input picture into its different channels and send it off to both local and global branches. Global branch uses 2D FFT to map the spatial domain feature to the frequency domain to model the global context of the image. Real and imaginary parts of frequency domain features make up a dual channel representation. They go through linear transformation, normalization, and nonlinear activation to improve cross-band feature interaction. Then an Inverse Fourier Transform is applied which transforms the frequency domain information back into the spatial domain and captures the long range dependency of the global semantic structure as well as making the missing region more coherent.

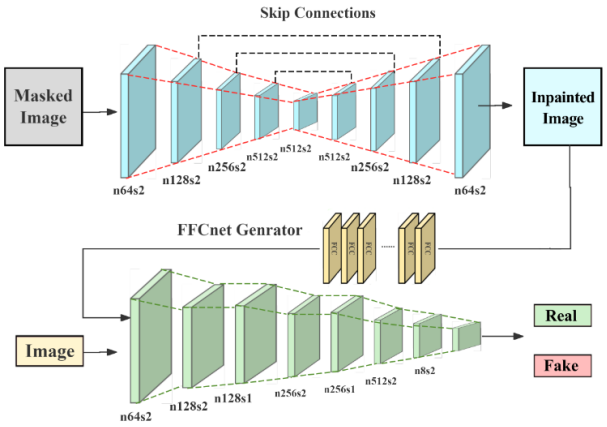


Fig. 3: Big-LaMA Dual-Branch Architecture

In order to improve the coordination of local and global features, a learnable gating mechanism is added to the model to regulate the flow of feature information between local and global adaptively. A light weight 1×1 convolution creates a spatial gate map which enables the fusion of cross branch features with weights. This way, global semantic info can picky boost local details without messing up how exact local parts are, so it makes the details in new pictures better match up when fixing big damaged ones.

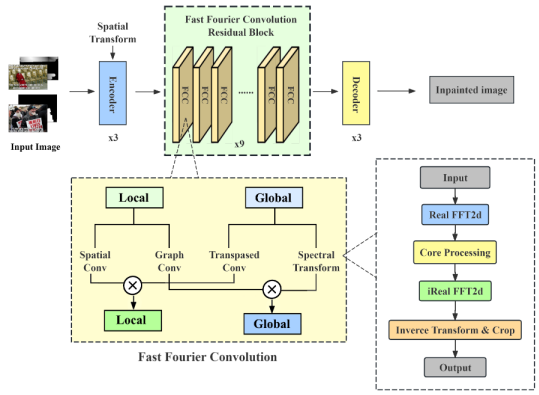


Fig. 4: Adaptive Feature Fusion with Spatial Gating

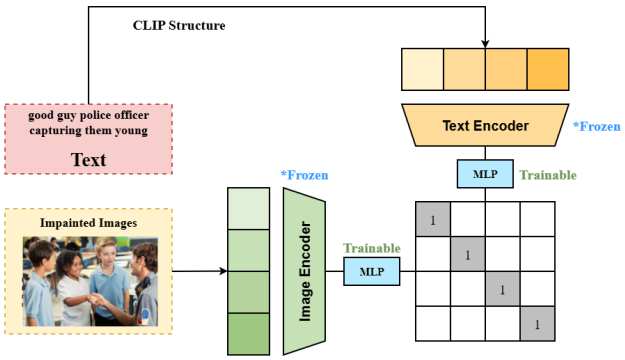


Fig. 6: The structure of CLIP model

After the feature fusion, the channel adaptation mechanism (Squeeze-and-Excitation module) further adjusts the features. The spatial information of every channel is pooled into descriptors by global average pooling, and then channel weights are created by a bottleneck fully connected network to scale the original features. This mechanism helps the model pay more attention to important channels so that it can use all kinds of big and small details from different parts of the picture at the same time. Therefore, it improves the texture accuracy and structure consistency in the restored areas.

After such preprocessing, the input image has a better ability to retain low-frequency structure information inside the generative network. At the same time, adding global semantic prior can generate natural and coherent images for large defects or complex texture problems. The results are shown as follows:



Fig. 5: Big-LaMA Inpainting Results Comparison

B. Cross-modal encoding based on CLIP

In hate meme detection tasks, both the image and the accompanying text typically carry essential, interdependent semantic information.

Their relationship is inherently multimodal and inseparable: analyzing either modality in isolation risks overlooking critical cross-modal cues. For instance, the same meme image, such as a seemingly neutral facial expression or a widely reused template, which can convey radically different meanings depending on the specific textual caption.

Thus, to achieve holistic and context-aware understanding of memes, this work leverages the **Contrastive Language–Image Pretraining(CLIP)** model to jointly encode images and text into a shared, aligned semantic embedding space.

This enables direct comparison and fusion of visual and linguistic representations, facilitating robust detection of subtle, compositionally emergent hate signals that arise only from their interaction. Therefore, we use the CLIP visual encoder $E_v(\cdot)$ and the text encoder $E_t(\cdot)$ to map the input to the same semantic space $\mathbb{R}^D$.

The visual encoder of CLIP is to encode the emoji images into global image vectors which show overall semantics and patch-level vectors which show local details; similarly, we use the text encoder of CLIP to encode the texts on the pictures into global text vectors and token-level vectors.

1) Image encoding

Given an input meme image $\mathbf{I} \in \mathbb{R}^{3\times H\times W}$, we extract multi-level visual representations using the CLIP visual encoder $E_v(\cdot)$. In practice, E_v (typically a ViT-style backbone) outputs a sequence of visual tokens, including one global token and P patch-level tokens:

$$\mathbf{Z}_v = E_v(\mathbf{I}) \in \mathbb{R}^{(P+1)\times d_v}, \tag{2}$$

$$\mathbf{h}_v = \mathrm{ImgPool}(\mathbf{Z}_v) = \mathbf{Z}_v^{[\mathrm{CLS}]} \in \mathbb{R}^{d_v}, \tag{3}$$

$$\mathbf{H}_v = \mathrm{PatchProj}(\mathbf{Z}_v) = \mathbf{Z}_v^{[1:P]} \in \mathbb{R}^{P\times d_v}, \tag{4}$$

where $\mathbf{h}_v$ is the global image embedding that summarizes the overall semantics of the meme image, and $\mathbf{H}_v$ consists of patch-wise embeddings that preserve fine-grained visual details (e.g., facial expressions, objects, symbols, or local attributes) which are often critical in hateful memes.

To adapt CLIP representations to hateful meme detection while preserving the general-purpose vision–language knowledge learned from large-scale pre-training, we keep E_v frozen and introduce a lightweight MLP projection head $g_v(\cdot)$ to map both global and local features into a unified D-dimensional space:

$$\hat{\mathbf{h}}_v = g_v(\mathbf{h}_v) \in \mathbb{R}^D, \tag{5}$$
$$\hat{\mathbf{H}}_v = g_v(\mathbf{H}_v) \in \mathbb{R}^{P\times D}, \tag{6}$$

where $g_v(\cdot)$ is applied row-wise for matrix inputs (i.e., to each patch embedding). A simple two-layer MLP is sufficient in our setting:

$$g_v(\mathbf{x}) = \mathbf{W}_2\,\sigma(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2, \tag{7}$$

with a non-linear activation $\sigma(\cdot)$.

Freezing E_v mitigates overfitting on small-scale meme datasets and avoids catastrophic forgetting of CLIP's robust visual semantics, while the projection head learns task-aligned feature calibration.

a) Normalization.

For similarity computation in the shared embedding space, we further ℓ_2-normalize the projected features:

$$\tilde{\mathbf{h}}_v = \frac{\hat{\mathbf{h}}_v}{\|\hat{\mathbf{h}}_v\|_2}, \qquad \tilde{\mathbf{H}}_v = \mathrm{row_norm}(\hat{\mathbf{H}}_v). \tag{8}$$

2) Text encoding

For the meme caption (including OCR-extracted text when applicable), we tokenize it as $\mathbf{T} = \{t_1, t_2, \ldots, t_L\}$ and feed it into the CLIP text encoder $E_t(\cdot)$ to obtain a sequence of textual token embeddings:

$$\mathbf{Z}_t = E_t(\mathbf{T}) \in \mathbb{R}^{(L+1)\times d_t}, \tag{9}$$
$$\mathbf{h}_t = \mathrm{TextPool}(\mathbf{Z}_t) \in \mathbb{R}^{d_t}, \tag{10}$$
$$\mathbf{H}_t = \mathrm{TokenProj}(\mathbf{Z}_t) \in \mathbb{R}^{L\times d_t}, \tag{11}$$

where $\mathbf{h}_t$ is a global text embedding (e.g., the end-of-text pooled representation) capturing the overall meaning, tone, and implied stance of the caption, while $\mathbf{H}_t$ contains token-level embeddings that preserve local linguistic cues such as negation, sarcasm triggers, group identifiers, or subtly offensive phrases.

Projection with frozen text encoder: Analogous to the image branch, we freeze E_t and employ an MLP projection head $g_t(\cdot)$ to map textual representations into the same D-dimensional space:

$$\hat{\mathbf{h}}_t = g_t(\mathbf{h}_t) \in \mathbb{R}^D, \tag{12}$$
$$\hat{\mathbf{H}}_t = g_t(\mathbf{H}_t) \in \mathbb{R}^{L\times D}. \tag{13}$$

We also apply ℓ_2 normalization:

$$\tilde{\mathbf{h}}_t = \frac{\hat{\mathbf{h}}_t}{\|\hat{\mathbf{h}}_t\|_2}, \qquad \tilde{\mathbf{H}}_t = \mathrm{row_norm}(\hat{\mathbf{H}}_t). \tag{14}$$

Patch–token semantic alignment: In the shared cross-modal space, we quantify local alignment between image patches and text tokens by cosine similarity. Specifically, we construct a patch–token affinity matrix:

$$\mathbf{A} = \tilde{\mathbf{H}}_v\tilde{\mathbf{H}}_t^\top \in \mathbb{R}^{P\times L}, \qquad A_{ij} = \cos(\hat{\mathbf{H}}_v^{(i)}, \hat{\mathbf{H}}_t^{(j)}), \tag{15}$$

which measures how strongly the i-th image patch semantically corresponds to the j-th text token.

This local cross-modal alignment is particularly useful for hateful meme understanding, as many hateful implications are revealed only when specific visual regions (e.g., identity cues, symbols, gestures) are interpreted together with specific words or phrases. The global similarity score can be computed similarly as $s = \tilde{\mathbf{h}}_v^\top \tilde{\mathbf{h}}_t$ when needed for coarse-level alignment.

C. Presupposed Context Module (PCM)

In multimodal hateful meme detection, a single modality is often insufficient to reveal the implicit hateful intent. A meme image may appear visually neutral when viewed in isolation, while the accompanying text can also be ambiguous or seemingly harmless when read alone.

However, hateful meaning frequently emerges only when the two modalities are interpreted together, where the text "presupposes" a specific visual context (or vice versa) and the joint reading activates the intended offensiveness.

Motivated by this observation, we propose a *Presupposed Context Module (PCM)*, whose goal is to explicitly model the semantic dependency between the global image and global text representations and thereby construct a stronger joint context for subsequent multimodal reasoning. The following diagram figure7 illustrates the general mechanism of PCM. The mechanism of PCM will be elaborated in detail in the following part of the article.

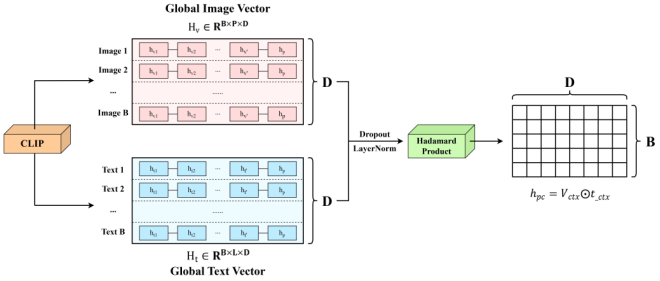


Fig. 7: Presupposed Context Module schematic diagram

For the input, we let $\mathbf{h}_v \in \mathbb{R}^D$ and $\mathbf{h}_t \in \mathbb{R}^D$ denote the global image and global text embeddings produced by the CLIP-based encoders and projection heads described in Section 4.2. PCM operates on these global vectors, focusing on coarse-grained semantic alignment that captures the overall stance and intent of the meme.

PCM first applies two modality-specific non-linear transformations to map $\mathbf{h}_v$ and $\mathbf{h}_t$ into a task-adaptive semantic space:

$$\tilde{\mathbf{h}}_v = f_v(\mathbf{h}_v), \tag{16}$$
$$\tilde{\mathbf{h}}_t = f_t(\mathbf{h}_t), \tag{17}$$

where $f_v(\cdot)$ and $f_t(\cdot)$ are lightweight MLP blocks.

Concretely, we implement each block as a stack of linear layers with GELU activation, followed by Dropout and LayerNorm:

$$f(\mathbf{x}) = \mathrm{LayerNorm}(\mathrm{Dropout}(\sigma(\mathbf{W}\mathbf{x} + \mathbf{b}))), \tag{18}$$

and apply the same form for both modalities (with independent parameters). This design improves robustness and encourages the projected features to be discriminative across dimensions while remaining stable during training.

To explicitly encode cross-modal interaction, PCM fuses the projected global representations using element-wise multiplication:

$$\mathbf{h}_{\mathrm{PCM}} = \tilde{\mathbf{h}}_v \odot \tilde{\mathbf{h}}_t, \tag{19}$$

where $\odot$ denotes the Hadamard product.

Compared to simple concatenation or additive fusion, the multiplicative interaction acts as a dimension-wise gating mechanism: only semantic dimensions that are simultaneously activated (i.e., consistent across modalities after projection) are emphasized, while uncorrelated or noisy dimensions are naturally suppressed. This property is well-suited to hateful memes, where the final meaning is often carried by a small set of aligned cues spanning both image and text.

In the PCM, the semantics of the image and the text are combined via the Hadamard product (element-wise multiplication). The main idea of this Hadamard product is to integrate information from the image and text into a joint representation through element-wise interaction.

When both the image and the text contain hateful information, their semantics after transformation will produce a relatively large h_{PC} value, and after ReLU activation, the hateful features become more prominent. If only the image contains hateful information, the semantics of the image will be strengthened while the semantics of the text will be weakened, and the resulting h_{PC} can still show hateful features. Similarly, if only the text contains hateful information, the semantics of the text will be strengthened while the semantics of the image will be weakened, and the resulting h_{PC} also shows hateful features. If neither the image nor the text contains hateful information, their semantics are close to zero, and after ReLU activation, h_{PC} outputs zero, and the model will not classify it as a hateful meme.

Optionally, we apply a non-linearity to highlight informative activations:

$$\bar{\mathbf{h}}_{\mathrm{PCM}} = \mathrm{ReLU}(\mathbf{h}_{\mathrm{PCM}}), \tag{20}$$

so that weak or contradictory interactions are dampened, while salient joint evidence remains prominent for downstream modules/classifiers.

Following the adaptation principle in Section 4.2, we keep the CLIP image and text encoders frozen and train only the lightweight PCM parameters. This strategy preserves CLIP's generalizable vision: language knowledge learned from large-scale pre-training and reduces overfitting risk on relatively small hateful meme datasets.

As a result, PCM concentrates on learning **task-specific cross-modal coupling** rather than modifying the foundational unimodal representations.

D. Semantic-Context Fusion Module (SCFM)

Although we have previously established the PCM (Presupposed Context Module), which can effectively capture explicit hate signals when both image and text are present by modeling their joint semantics, PCM still has limitations. It relies solely on the direct representations of image and text, making it difficult to identify underlying sarcasm, implicit accusations, or factual distortions—these require social common sense and language understanding.

To address this, we designed the Semantic-Context Fusion Module (SCFM), which leverages large language models (LLM) to provide high-level semantic guidance. This allows the model to focus on potential misleading or implicit hateful information. The problem is that while CLIP understands the content, it doesn't necessarily grasp the subtext or distorted facts. SCFM consists of two complementary submodules:

1. **Semantic Guidance Module (SGM)**: It uses LLM to provide high-level semantic priors, helping the model identify potential misleading or hateful information in both text and images, and weight their importance.

2. **Graph-based Multimodal Module (GMM)**: Based on the results from SGM, this module models the complex interaction between text tokens and image patches, allowing each element's representation to integrate cross-modal contextual information.

Through these two components, SCFM compensates for PCM's shortcomings in implicit semantic understanding, enabling the model to not only grasp surface-level information but also reason about potential sarcasm, allusion, or factual distortion, thereby improving its ability to identify hateful memes more accurately.

1) Semantic Guidance Module (SGM)

In the task of hate meme detection, much of the hateful or misleading information often lies in higher-level semantic relationships, such as exaggeration, fabrication, rumors, or false causality. The question of "Are there are false claims?" is a strong intermediate semantic signal, but these dangers are often subtle and not easily understood by models that rely solely on local features.

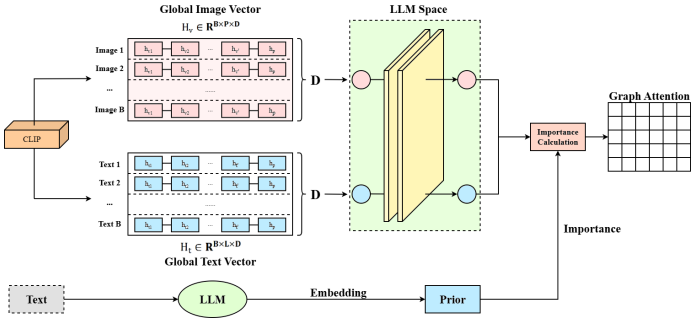


Fig. 8: The structure of Semantic Guidance Module (SGM)

Hate memes often rely on false facts, implicit accusations, stereotypes, and sarcasm, elements that are not explicitly stated

in the image or text. Instead, they are hidden in "social consensus" and "common knowledge," areas where large language models (LLMs) excel in capturing and understanding these underlying semantic cues.

To help the model better grasp these subtle signals, we designed the Semantic Guidance Module (SGM). Its core idea is to leverage the semantic priors provided by LLMs to guide the model's attention towards local content that may carry significant information, forming a semantically-guided multimodal representation.You can see the structure of it in Figure.11.

We transform the current meme into an abstract semantic question, such as "Are there any false claims? Are these hateful content?" to guide the model's attention towards the semantic prior.

Using the text extracted from the meme through OCR, and by leveraging the embedding vectors of large language models (LLMs), we obtain a high-level semantic vector as prior information. This semantic prior is used to assess the importance of different image regions (patches) and text tokens in terms of potential misinformation, accusations, or implications. Based on this importance evaluation, we reinforce the relevant tokens while suppressing the irrelevant ones, thus enhancing the model's sensitivity to key information.

Specifically, given the image patch representation $H_v \in \mathbb{R}^{B\times P\times D}$ and the text token representation $H_t \in \mathbb{R}^{B\times T\times D}$, the SGM first maps them into a unified semantic space:

$$\tilde{H}_v = \text{Proj}_{\text{LLM}}(H_v), \quad \tilde{H}_t = \text{Proj}_{\text{LLM}}(H_t) \tag{21}$$

The projected vectors of the text and image are then concatenated along the token dimension to form a multimodal token sequence $M = [\tilde{H}_t; \tilde{H}_v] \in \mathbb{R}^{B\times N\times d}$, where $N = T+P$.

Next, the SGM utilizes an LLM to generate a semantic vector $p \in \mathbb{R}^{B\times d}$ based on a task-specific prompt (In our task, we use "Are there false claims?"). The response strength of each token to the semantic prior is computed via normalization and cosine similarity:

$$\alpha_{b,n} = \frac{\exp(\tilde{m}_{b,n}\cdot p_b)}{\sum_{n'}\exp(\tilde{m}_{b,n'}\cdot p_b)}, \quad \tilde{m}_{b,n} = \frac{m_{b,n}}{\|m_{b,n}\|} \tag{22}$$

These response coefficients are used to adjust the representation of multimodal tokens:

$$\hat{m}_{b,n} = m_{b,n} + \alpha_{b,n}\cdot m_{b,n}, \quad \hat{M} = \text{TokenFuse}(\hat{m}) \tag{23}$$

Finally, the global semantic perception vector h_{sp} is generated through weighted pooling:

$$h_{sp} = \text{Pool}\left(\sum_{n=1}^{N}\alpha_n\hat{m}_n\right) \tag{24}$$

By letting the model refer to the LLM's prior, SGM can guide the alignment of image–text features in a more meaningful direction. This is especially important when dealing with hateful content that is hard to judge from pixels or a single sentence alone.

This process not only preserves the local details of images and text but also injects the high-level semantic prior provided by the LLM, making the model more semantically sensitive when detecting hate or misleading information. When a meme contains subtle false cues, text puns, or metaphorical patterns, SGM helps the model to understand these signals.

2) Graph-based Multimodal Module (GMM)

In the last section, we mentioned that SGM evaluates which tokens in the image and text are worth focusing on by introducing semantic priors, helping the model identify the important parts related to the task. It mainly focuses on the importance evaluation of tokens, but it does not model the interactions between these tokens. Therefore, we need to establish a module to model the relationships between them.

Therefore, we built the Graph-based Multimodal Module (GMM), a cross-modal reasoning module based on graph neural networks (GNNs). Its core idea is to treat text tokens and image patches as a multimodal graph and perform message passing on this graph to achieve cross-modal contextual understanding.

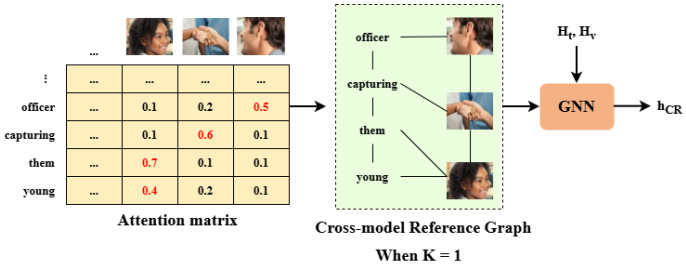


Fig. 9: The structure of Graph-based Multimodal Module

We construct three types of graph edges: first, intra-text relations, which capture the sequential relationships between text tokens; second, intra-image relations, which capture the spatial adjacency between image patches; and third, cross-modal relations, which are formed by highly relevant connections provided by SGM, helping the model capture strong links between image and text.

During the graph construction process, we first use SGM to capture the cross-modal relationships by computing a correlation matrix A between text and image. Using the semantic vectors of the image and text provided by SGM, we compute a similarity matrix:

$$A_{b,n,m} = \frac{\langle \mathbf{s}_{b,n}, \mathbf{s}_{b,m}\rangle}{\sqrt{d}} \times \text{importance}_{b,n,m} \tag{25}$$

In the previous section, we mentioned that to enhance the model's comprehension capability, we also introduced a social perception vector. This vector is generated via a weighted pooling method that aggregates information from each token. Specifically, we weight the importance of each token and then perform a pooling operation.

After constructing the graph and generating the semantic-aware vectors, we use a graph neural network (GNN) for message passing. Through the GNN, information can be propagated between the nodes of the graph, so that each node not only contains its own information but also receives information from its neighboring nodes. For example, image patches and text tokens are connected by edges and exchange information through the GNN layers:

$$\begin{aligned} x &= \text{GNN1}(x, \text{edge_index}) \\ x &= \text{GNN2}(x, \text{edge_index}) \end{aligned} \tag{26}$$

After message passing through the GNN layers, we aggregate the representation of each node to obtain a final cross-modal semantic vector. This vector contains the combined information from both the image and the text.

$$h_{\text{CR}} = \frac{1}{B} \sum_{i=1}^{B} x_i \tag{27}$$

We transform text tokens and image patches into a unified graph structure and use a GNN for message passing, so that each token and patch carries cross-modal contextual information.

This information propagation not only enhances the semantic representation of each token but also promotes deep fusion between image and text. As a result, each token is no longer isolated but possesses a semantic representation informed by cross-modal context.

E. Final Classifier

In this model, the final classifier integrates the outputs from the PCM, SPM, and CRM modules. The output of the PCM module, h_{PC}, the output of the SPM module, h_{sp}, and the output of the CRM module, h_{cr}, represent important information from different aspects of the multimodal data.

The whole process is as following Figure:

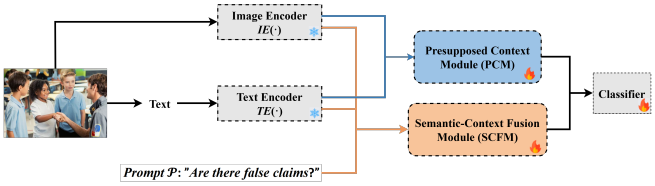


Fig. 10: The whole process of HateReasoner

The PCM module combines image and text features to provide a joint contextual representation, h_{PC}, capturing the interaction between these two modalities. The SPM module generates a semantic prior vector, h_{sp}, which is produced using an external language model and provides additional semantic context for the model. Finally, the CRM module outputs h_{cr}, which contains the cross-modal reasoning results between the image and text.

After concatenating these three vectors, h_{PC}, h_{sp}, and h_{cr}, they form a unified feature vector. This vector contains all the important information from the image, text, and semantic prior.

The concatenated vector is then passed into a fully connected layer (classifier), which outputs the final prediction logits. These logits represent the unactivated scores for each sample, indicating the model's confidence in whether the sample belongs to the positive class.

The formula is as follows:

$$\text{logits} = \text{classifier}(h_{PC}, h_{sp}, h_{cr}) \tag{28}$$

To train the model, the loss function needs to measure the difference between the predicted logits and the true labels. We use Binary Cross-Entropy Loss (BCEWithLogitsLoss), which is a common loss function for binary classification tasks and is suitable when the model outputs raw logits (i.e., the outputs before applying the Sigmoid activation).

This loss function computes the error between the predicted logits and the true labels, and uses it for backpropagation to optimize the model parameters.

The formula is as follows:

$$\text{Loss} = \text{BCEWithLogitsLoss}(\text{logits}, \text{labels}) \tag{29}$$

In the forward pass of the model, the logits obtained need to go through the Sigmoid activation function, which converts them into probability values for the final binary classification prediction. The Sigmoid function maps the logits to a range of [0, 1], representing the probability that the sample belongs to the positive class.

IV. EXPERIMENTAL RESULTS

A. Dataset

The primary dataset used in our evaluation is the Hateful Memes Challenge (HMC) dataset (8), from the Hateful Memes Challenge.



Fig. 11: Example of Hateful Memes Challenge Dataset

This dataset contains 10,000 meme images, each with a corresponding textual description, forming 10,000 multimodal

samples. The dataset is divided into 8,500 training meme images, 500 development images (seen splits), 540 development images (unseen splits), 1,000 test images (seen splits), and 2,000 test images (unseen splits). We split the training set into 70%, validation set into 10%, and test set into 20%.

B. Hyperparameter Setup

Hyperparameter	HateReasoner
Image size	224
Pretrained CLIP model	ViT-Large-Patch14
Projection dimension (n)	768
Pre-output dimension (m)	1024
Optimizer	AdamW
Maximum epochs	20
Batch size	32
Learning rate	1e-5
Weight decay	1e-4
Gradient clip value	0.1

TABLE I: Hyperparameter Settings for HateReasoner

The hyperparameter configuration for our HateReasoner model employs a standard set of training specifications, as summarized in Table I .

The model is built upon a ViT-Large-Patch14 CLIP backbone, processes images at a resolution of 224×224, and utilizes the AdamW optimizer with a learning rate of 1e-5. Training is conducted for 20 epochs with a batch size of 32, a weight decay of 1e-4, and a gradient clipping value of 0.1, ensuring stable and efficient optimization.

C. Evaluation Metrics

For evaluation metrics, we chose AUROC and accuracy.

AUROC is a common metric for assessing the performance of classification models. It measures the model's overall performance across different decision thresholds, not just at a specific threshold. It reflects the model's ability to distinguish between positive and negative samples.

In the task of hateful meme detection, accuracy is important because it provides a clear indication of the model's overall correctness. A higher accuracy suggests that the model effectively classifies the majority of data correctly, especially when the goal is to distinguish between hateful and non-hateful content. This metric directly shows the model's ability to predict the target class.

Type	Model	Acc. (%)	AUROC (%)
	Human annotators (9)	84.70	82.65
Unimodal	Image-grid(10)	52.00	52.63
	Image-region(11)	52.13	55.92
	Text BERT(12)	59.20	65.08
Multimodal	Late fusion	59.66	64.75
	Concat BERT(11)	59.13	65.79
	MMBT-grid(11)	60.06	67.92
	MMBT-region(11)	60.23	70.73
	ViLBERT CC(11)	61.10	70.03
	Visual BERT COCO(11)	64.73	71.41
	GPT-4o mini(13)	69.50	75.02
Our Method	**HateReasoner**	**73.40**	**77.99**

TABLE II: Performance Comparison of Different Models

Quantitative Analysis In this study, we evaluated the performance of various unimodal and multimodal methods for hate speech detection and compared them to human annotators.

The results show that pure text models, such as BERT, and early multimodal fusion methods, like Late Fusion and Concat BERT, perform relatively poorly. Their accuracy and AUROC are significantly lower than the human baseline (84.70%, 82.65%).

On the other hand, advanced multimodal models that combine both visual and textual information, such as ViLBERT and Visual BERT, demonstrated stronger detection capabilities. Among them, the large-scale pre-trained vision-language model GPT-4o mini achieved the best results so far (69.50%, 75.02%).

Our proposed model, HateReasoner, further improves performance, reaching 72.40% in accuracy and 77.99% in AUROC. This is a significant improvement over existing automated methods and brings us closer to human-level performance.

V. CONCLUSION

In this paper, we introduce HateReasoner, a novel multimodal framework for harmful meme detection, designed to address the challenge of recognizing context-dependent implicit harmful information generated by the combination of images and text. Unlike traditional methods that mainly rely on shallow feature alignment, our framework performs joint reasoning through three complementary semantic paths to capture complex hate signals.

First, the Presupposed Context Module (PCM) explicitly models the strong interactions between global image and text representations, creating a joint contextual understanding that emerges only when both modalities are combined. Next, the Semantic Context Fusion Module (SCFM) integrates high-level semantic guidance and structured relational reasoning. Its Semantic Guidance Module (SGM) leverages the commonsense and contextual knowledge from large language models to identify subtle cues such as irony, misinformation, and implicit accusations. Then, the Graph-based Multimodal Module (GMM) uses a graph neural network to model complex

relationships between local image patches and text tokens, enabling deeper semantic fusion and cross-modal reasoning.

Future work will focus on further enhancing the Semantic Guidance Module. Specifically, we plan to fine-tune the underlying large language model on domain-specific multimodal hate speech data, aiming to provide more accurate and nuanced semantic priors and thereby strengthen its capability as a knowledge provider. Additionally, we intend to explore extending the module to handle animated content, such as GIFs, enabling the detection of harmful or hateful elements in dynamic visual media (14).

Ultimately, we aim to integrate this technology into a practical system. Our goal is to develop a platform that allows users to freely input text, images, and animated content to generate memes, with HateReasoner serving as the core backend moderation module. The system would automatically detect and filter harmful content before posting, promoting creative expression while maintaining a healthy and safe online environment.

REFERENCES

[1] J. Akram, T. H. Siddiqui, and A. A. Akram, "Exposure to depression memes on social media increases depressive mood: Investigating the roles of rumination, emotional contagion, and moderating factors," *Frontiers in Psychology*, vol. 13, 2022.

[2] K. Müller and C. Schwarz, "From hashtag to hate crime: Twitter and anti-minority sentiment," *American Economic Review*, vol. 111, no. 11, pp. 3660–3699, 2021.

[3] G. K. Kumar and K. Nandakumar, "Hate-clipper: Multimodal hateful meme classification based on cross-modal interaction of clip features," *arXiv preprint arXiv:2210.05916*, 2022. [Online]. Available: https://arxiv.org/abs/2210.05916

[4] R. Suvorov, D. Ulyanov, and E. Logacheva, "Big lama: Scalable and high-quality image inpainting," *arXiv preprint arXiv:2304.06790*, 2023.

[5] R. Zhu *et al.*, "Enhance multimodal transformer with external label and in-domain pretrain: Hateful meme challenge winning solution," *arXiv preprint arXiv:2012.08290*, 2020. [Online]. Available: https://arxiv.org/abs/2012.08290

[6] K. Li, Y. Zhang, K. Li, Y. Li, and Y. Fu, "Visual semantic reasoning for image-text matching," in *European Conference on Computer Vision (ECCV)*, 2019.

[7] X. Zhou, C. Yao, H. Wen, Y. Wang, S. Zhou, W. He, and X. Liang, "East: An efficient and accurate scene text detector," in *CVPR*, 2017.

[8] D. Kiela, H. Firooz, A. Mohan, V. Goswami, A. Singh, S. Mazar'e, D. Ju, Y. Hua, W. Galuba *et al.*, "Detecting hate speech in multimodal memes," in *NeurIPS 2020 Competition and Demonstration Track*, 2020.

[9] A. Anaissi, J. Akram, K. Chaturvedi, and A. Braytee, "Detecting and understanding hateful contents in memes through captioning and visual question-answering," *arXiv preprint arXiv:2504.16723*, 2025. [Online]. Available: https://arxiv.org/pdf/2504.16723

[10] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2015.

[11] D. Kiela, H. Firooz, A. Mohan, V. Goswami, A. Singh *et al.*, "The hateful memes challenge: Competition report," in *Proceedings of the NeurIPS 2020 Competition and Demonstration Track*, 2021, pp. 344–360.

[12] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," in *NAACL*, 2019.

[13] OpenAI, "Gpt-4o: Openai's most advanced multimodal model," 2024, accessed: 2024-10-03. [Online]. Available: https://openai.com/index/gpt-4o

[14] Q. Sun, T. Chen, Y. Zhang, Y. Zhang, J. Yue, J. Jiao, and Z. Fu, "Multihateloc: Towards temporal localisation of multimodal hate content in online videos," *arXiv preprint arXiv:2512.10408*, 2025, accessed: 2025-12-17. [Online]. Available: https://doi.org/10.48550/arXiv.2512.10408